

prog01v1:

My program starts off with accepting 2 arguments from the command-line. This makes it so any valid path, absolute or relative, can be accepted into the program. I use Fopen to open the input and output files. I store the first line of the program, n, as an integer, so I can use it to iterate through a for-loop.. I use the loop with fscanf to read pairs of floats, x and y, so that the decimal gets stored as well. I used malloc for storing the x points and y points. This is because I would not know how big the arrays are until runtime, so the amount of space that gets allocated is dependent on the input. I created findminmax as a helper function to make the code easier to understand and debug. I also have to find the min and max values of the arrays multiple times, so putting this piece of code into a separate function reduces the need for repeating code. Before printing, I use floor and ceil to round the upper and lower bounds. By using $\ast 10 / 10.0$, the float stays with one decimal place.

Prog01v2:

v2 is very similar to v1 with some changes. My program now takes multiple inputs. I loop through arv[] and read each file. Now, instead of storing all of the x and y values into an array, I just look at one x and y pair at a time and determine if it is bigger/smaller than the current max/min. All the other logic is the same.

[build.sh](#):

Compiled with -wall and -std=c11

Script01_1.sh

Stored input and output file paths into variables

Script01_2.sh

Used \$@, which stores stores the full list of arguments

Script01_3.sh.

The while loop reads each line in the input file and stores it into the inputFiles list